

NEW PRODUCTS

MAKE IN INDIA

COMPONENTS

Supercapacitors for automotive use

FMU series high-performance supercapacitors for automotive electronics from KEMET use a proprietary



aqueous electrolyte solution that provides high durability against liquid leakage, vibrations, and thermal shock,

which improves its reliability even in harsh automotive environments. The capacitors are highly reliable in a wide operational temperature of -40 to +105°C. These miniature supercapacitors deliver extended lifetime, high power density, fast charging, and high reliability for various energy storage applications. The FMU supercapacitors are ideal for automotive applications that require a main power system backup during a power loss, such as ADAS, autonomous vehicles, and central gateway ECUs. These can be employed in equipment ranging from IoT devices, smart meters, medical devices, and industrial computing.

Kemet

<https://www.kemet.com>

Antennas supporting Wi-Fi 7

Antenova has released a trio of high-efficiency antennas for Wi-Fi 6 and Wi-Fi 6E devices that will offer support for the upcoming Wi-Fi 7 standard. The antennas include a surface mount antenna named Billi



Billi SR43W078

Lotti SRF3W077

Nitida SRE3W084

SR43W078, a flexible antenna called Lotti SRF3W077, and an external antenna called Nitida SRE3W084. All three antennas have high efficiency and are simple to add to a product since they don't require a matching network. All three use the 2.4GHz, 5GHz, and 6GHz bands and support IEEE standards 802.11a/b/g/j/n/ac/ax. These antennas will deliver better online experiences and support simultaneous communications to many endpoints with increased security. The antennas will enable users to play high-quality content, such as supporting 4K and 8K video streaming, networked gaming, video calling, etc, without any lag, using wireless connectivity. They can be employed in gaming consoles, set-top boxes, surveillance cameras, networked IoT devices, etc.

Antenova

<https://www.antenova.com>

Sensor for surveillance cameras

Omnivision OSO4D is a 2K complementary metal oxide semiconductor (CMOS) image sensor for consumer security and surveillance applications. The camera sensor comes in an



8.4mm optical format and has an image resolution of 4 megapixels. It features a 2.0-micron, high-performance backside illuminated pixel, enabling the camera to capture images with higher sensitivity, ultra-low noise, and better overall image quality. The OSO4D image sensor is power-efficient and offers an improvement of over 30% in signal-to-noise ratio when

compared with its predecessor. It will enable designers to provide security with better video quality and long-lasting battery run time. Its higher frame rate will enable it to capture high-quality videos and still images of fast-moving objects. Furthermore, it will enable security cameras to have clearer images in darker environments. The image sensor is suitable for use in both analogue and IP cameras, and can be used in cameras meant for smart home surveillance, doorbells, and baby monitoring devices.

Omnivision

<https://www.ovt.com>

Sound driver IC

PAM8906 from Diodes Incorporated is a new driver IC that enables designers to improve the output from piezoelectric sounders. The PAM8906 can produce louder sounds that can be



carried over longer distances by integrating a synchronous boost converter in the IC.

This driver also integrates functionality that prolongs its battery-powered operation. The sound driver operates with either an external pulse width modulation (PWM) input or in self-excitation mode, thus providing greater application flexibility and allowing support for an increased variety of sounders. The PAM8906 has three output voltage variants: 20VPP, 24VPP, and 36VPP. This enables the creation of different output versions of the same system. It will enable designers to build low-power compact devices with high volumes. The PAM8906 driver IC is suitable for use in smoke/gas/water alarms, air humidifiers, handheld GPS devices, industrial security alarms, medical